

## PORTFOLIO

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# 01 Core Strengths - Highlights from Flagship Projects

Sole engineer on the AI chat backend

**OSS → Production**

From openCHA research framework to a deployed production agent in 5 months

Agentic AI Chat Task Completion Rate Improvement

**55% → 84%**

Designed and ran 25-scenario multilingual benchmark  
Improvements made based on benchmark findings

Time Series Based ML Air Pollution Prediction

**73 – 95%**

Average error reduction of the best model (LSTM) over its baseline.

## ● Used in production

Python · FastAPI (REST + SSE) · PostgreSQL · async SQLAlchemy · LangChain SQL agent · Gemini API · Docker · GCP · NGINX · Prompt Engineering · Agentic Systems

## ● Used in research

PyTorch · LSTM / 1DCNN / GRU · transfer learning · time-series forecasting · benchmark & rubric design

## ● Working knowledge

RAG (LangChain, HuggingFace embeddings) · C / C++ · Arduino · Raspberry Pi · PIC

# 02 TripDoc AI Chat Assistant

TripDoc (medical tourism startup) · Sole AI Engineer · Jan – Jun 2026

## Purpose

A multilingual chat assistant embedded in TripDoc's existing medical-tourism app. Users search hospitals, pharmacies, treatments and products near a location, get general health guidance, handle emergency queries, and complete a booking, all inside one natural-language conversation.

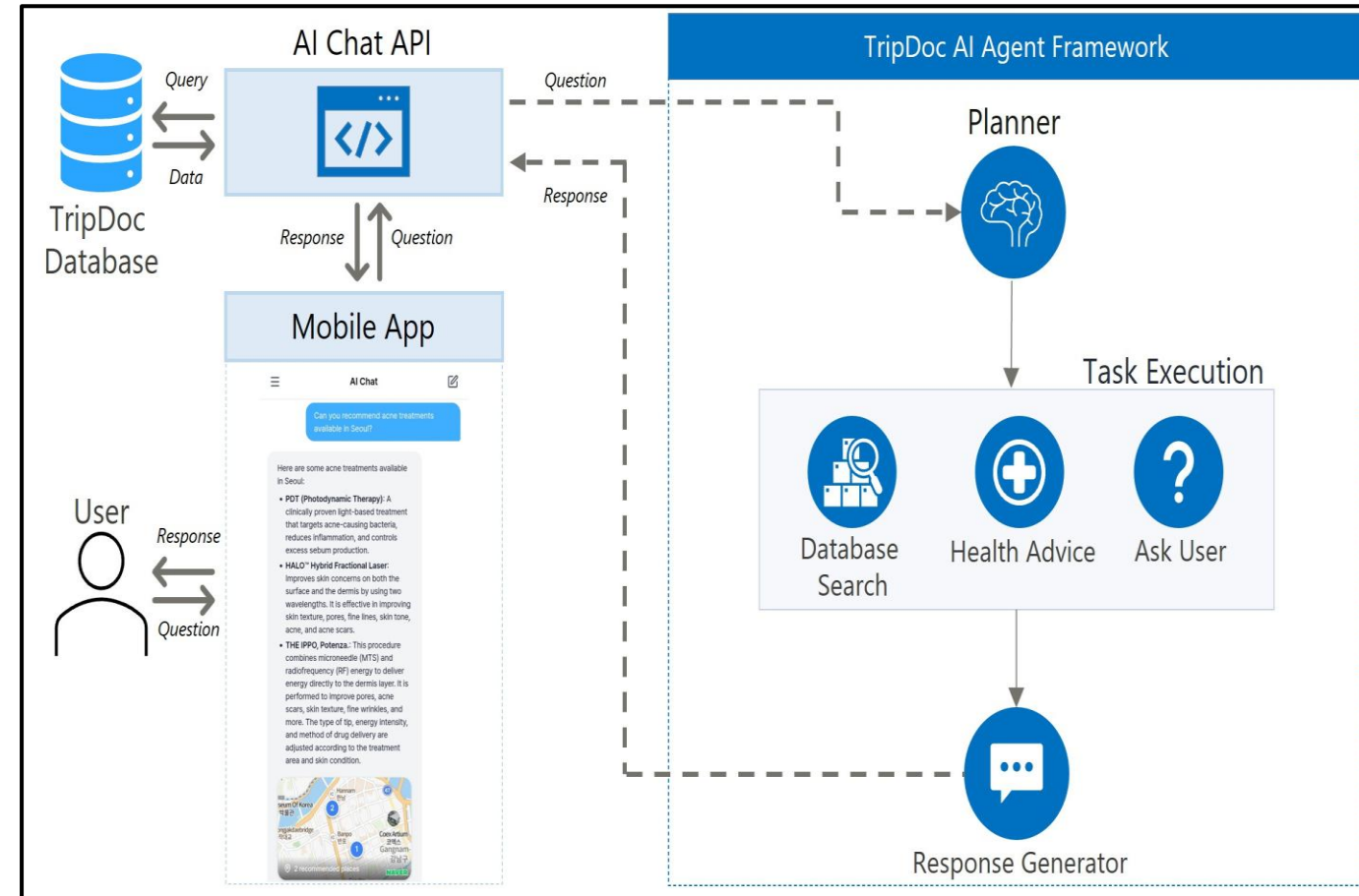
## Scope of Work

### Starting point:

- openCHA (UC Irvine research framework): orchestrator / planner / task / response-generator skeleton.
- Existing systems: mobile app, chat UI, and TripDoc database

### Built:

- Multi-stage planner — conversation-state extraction, intent classification, plan generation, with intent-based prompt routing
- TripDoc DB query agent – multilingual data handling with responses in the user's language
- Multi-turn booking flow and its deterministic stage controller
- Emergency situation handling and routing to emergency rooms
- Production API layer — session store, SSE streaming, concurrency control, Docker/GCP deployment
- Competitive benchmark design, execution and analysis
- End-to-end system architecture documentation



High level AI Chat Architecture Diagram

**Turn 1** – Entities resolved from the TripDoc DB and ID-validated leading to clickable product links + map showing location of the related hospitals

AI Chat

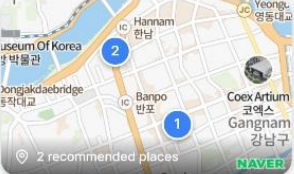
Recommend acne treatments in Seoul

Hello! There are several effective acne treatments available in Seoul that target different skin concerns. Here are the options currently available:

- **PDT (Photodynamic Therapy)** at **Cham Jin Han Clinic** (43 Nonhyeon-ro 85-gil, Gangnam District): A clinically proven, light-based treatment that targets acne-causing bacteria, reduces inflammation, and controls excess sebum production.
- **HALO™ Hybrid Fractional Laser** at **Cham Jin Han Clinic** (43 Nonhyeon-ro 85-gil, Gangnam District): This treatment uses two wavelengths to improve skin texture, pores, acne, and acne scars.
- **THE IPPO, Potenza** at **The Pretty Clinic** (Naruteo-ro, Seocho District): This procedure combines microneedle (MTS) and radiofrequency (RF) energy to deliver energy directly into the dermis, helping to improve pores, acne scars, skin texture, and fine wrinkles.

To help me find the best clinic for you, could you tell me more about your acne concerns? For example, are you looking for basic skincare, laser treatments, or consultation for a specific skin condition?

Would you like to book a reservation for any of these treatments?



 **PDT (Photodynamic Th...**  
Cham Jin Han Clinic  
USD 134.77

 **HALO™ Hybrid Fraction...**  
Cham Jin Han Clinic  
USD 1,502.32

**Turn 2** – Booking stage entered; agent requests the missing required field (date/time)

I would like to book the Potenza treatment.

It is great that you have decided to proceed with the Potenza treatment at The Pretty Clinic! To get your appointment set up, please note the clinic's working hours: Monday-Friday 10:00-20:00; Saturday-Sunday 11:00-17:00. Please be aware that the 12:00-13:00 lunch break is excluded from these hours, and reservations are available every 30 minutes. What date and time would you prefer for your visit?

**Turn 3** – Clinic and treatment carried across turns by the stage controller; options and pricing pulled live from the DB

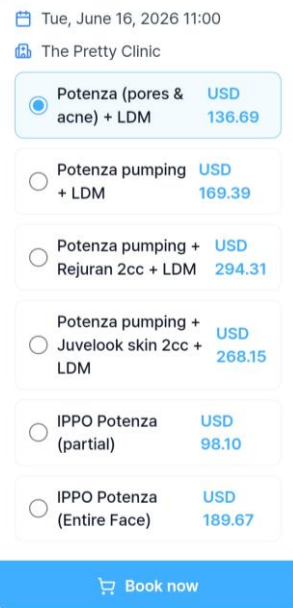
Tomorrow at 11am

I have noted your request for an appointment tomorrow, June 16, 2026, at 11:00 at The Pretty Clinic.

Here are the available options for the Potenza treatment:

1. Potenza (pores and acne) + LDM: Potenza (pores and acne) + LDM
2. Potenza pumping + LDM: Potenza pumping + LDM
3. Potenza pumping + Rejuran 2cc + LDM: Potenza pumping + Rejuran 2cc + LDM
4. Potenza pumping + Juvelook skin 2cc + LDM: Potenza pumping + Juvelook skin 2cc + LDM
5. IPPO Power (partial): IPPO Power (partial)
6. IPPO Power (Entire Face): IPPO Power (Entire Face)

Please let me know which of these options you would like to choose for your appointment, or if you would like more details about any of them.



**Turn 4** – Confirmation of selected option and custom booking link generated

Give me option 1

You have successfully selected the Potenza (pores & acne) + LDM treatment at The Pretty Clinic for June 16, 2026, at 11:00. You can now complete your reservation using the booking option provided. Would you like to book any other treatments?

 Tue, June 16, 2026 11:00

 The Pretty Clinic

**Potenza (pores & acne) + LDM** **USD 136.69**

 Book now

**Turn 3** – Context carried across turns regardless of language

내일 2시

**Turn 4** – Confirmation of selected option and identical custom booking link generated

1번 옵션 위

**Turn 2** – Same booking pipeline as English flow

여드름 BASIC 예약하고 싶어

메디큐브의원 예약 안내를 도와드리겠습니다. 진료 시간은 다음과 같습니다.

- 월요일: 10:30-13:00, 14:00-20:00
- 화요일-수요일: 10:30-13:00, 14:00-19:00
- 목요일-금요일: 10:30-13:00, 14:00-20:00
- 토요일: 10:30-16:00 (일요일은 휴무입니다)

오후 12:00-13:00은 점심시간으로 예약이 불가능하며, 예약은 30분 단위로 가능합니다. 원하시는 예약 날짜와 시간을 말씀해 주시겠어요?

8월 24일 오후 2시로 예약을 진행하겠습니다. 선택하신 메디큐브의원에서 제공하는 여드름 치료 옵션은 다음과 같습니다. 원하시는 옵션을 하나 선택해 주세요.

1. **여드름 BASIC**: 테라 또는 카프리, 재생라이트, 모델링팩이 포함되며, 필요 시 스티머와 전제 압출, 염증 주사가 진행됩니다.
2. **여드름 STANDARD**: 스티머, 테라, 카프리, 재생라이트, 모델링팩이 포함되며, 필요 시 스티머와 전제 압출, 염증 주사가 진행됩니다.
3. **여드름+붉은흉터 PLATINUM**: 스티머, 테라, 카프리, 시너지, 재생라이트, 모델링팩이 포함되며, 필요 시 스티머와 전제 압출, 염증 주사가 진행됩니다.
4. **여드름+붉은흉터+피지분비 PREMIUM**: 스티머, 테라, 카프리, 시너지, 피지다운주사, 재생라이트, 모델링팩이 포함되며, 필요 시 스티머와 전제 압출, 염증 주사가 진행됩니다.

위 옵션 중 어떤 것으로 예약을 도와드릴까요?

2026년 8월 24일 월 14:00

메디큐브의원

☒ 여드름 BASIC KRW 380,000

☐ 여드름 STANDARD KRW 490,000

☐ 여드름+붉은흉터 PLATINUM KRW 700,000

☐ 여드름+붉은흉터+피지분비PREMIUM KRW 900,000

바로예약

메디큐브의원에서의 2026년 8월 24일 오후 2시 '여드름 BASIC' 프로그램 예약이 선택되었습니다.

선택하신 1번 옵션으로 예약 과정이 완료되었습니다. 인터페이스에 제공된 예약 옵션을 사용하여 최종적으로 예약을 확정해 주시기 바랍니다.

2026년 8월 24일 월 14:00

메디큐브의원

여드름 BASIC KRW 380,000

바로예약

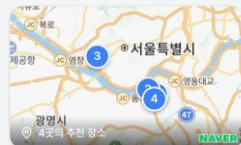
AI 채팅

여드름이 심한데 서울에서 치료법을 추천해 줘

서울에서 심한 여드름 및 관련 흉터 개선을 위해 고려해 보실 수 있는 치료법과 병원을 정리해 드립니다.

- **메디큐브의원 (서울 서초구)**
  - 치료법: 여드름 BASIC
  - 내용: 과다한 피지 분비 조절부터 노폐물 제거, 꼼꼼한 압출, 재생 라이트 및 진정 팩까지 포함된 토크스 케어입니다. (테라클리어 또는 카프리 레이저 선택)
- **강남오엘의원 (서울 강남구)**
  - 치료법: 리투오(Re2O) 스킨부스터
  - 내용: 피부결 개선에 효과적인 콜라겐 스킨부스터로, 여드름 흉터 개선에도 도움을 줄 수 있습니다.
- **더이베의원 (서울 서초구)**
  - 치료법: 더이베 포텐자 / 더이베 쥬베룩
  - 내용: 포텐자는 마이크로니들과 고주파를 이용해 모공과 여드름 자국을 개선하며, 쥬베룩은 자가 콜라겐 재생을 촉진해 패인 여드름 흉터에 효과적입니다.
- **다시불날의원 신촌홍대점 (서울 서대문구)**
  - 치료법: 포텐자 및 베크 니들 고주파
  - 내용: 미세한 니들을 통해 진피층에 고주파 에너지를 전달하여 피부 재생을 유도하며, 여드름 흉터 및 모공 개선에 도움을 줍니다.

구체적으로 레이저 시술이나 약을 치료 등 선호하시는 치료 방향이 있으신가요? 혹은 피부 상태 (예: 예민함 정도 등)에 대해 더 자세히 알려주시면, 상황에 가장 적합한 치료법을 선택하시는 데 추가적인 도움을 드릴 수 있습니다. 위 치료법 중 상담이나 예약을 원하시는 곳이 있다면 말씀해 주세요.



**더이베 쥬베룩 모음**  
더이베의원  
KRW 160,000

**리투오 (Re2O) 스킨부스터**  
강남오엘의원  
KRW 450,000

**여드름 BASIC**  
메디큐브의원  
KRW 470,000  
KRW 380,000

**더이베 포텐자**  
더이베의원  
KRW 150,000

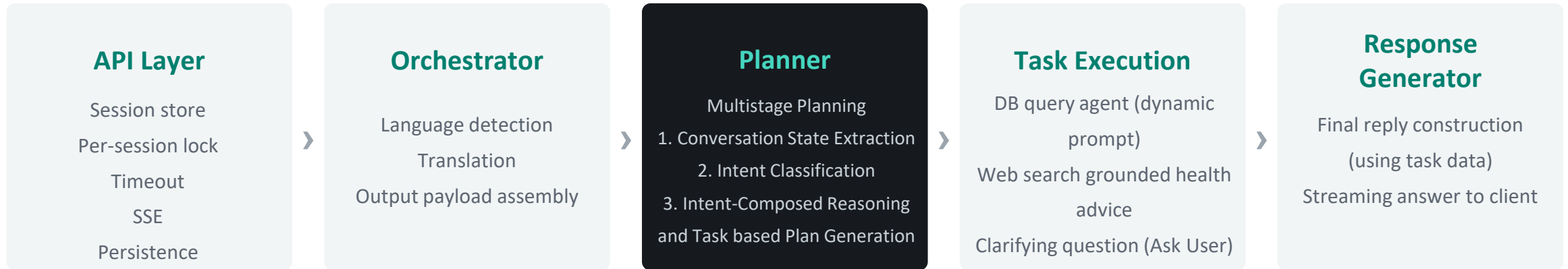
**Turn 1** – Korean Entities resolved from the TripDoc DB and ID-validated leading to clickable product links

**Note:** Product catalogue can vary by language in the DB – the agent surfaces what exists in the language rather than just translating English records

## 05

# System Architecture

Component structure follows the openCHA skeleton; each stage was rewritten or substantially extended for production.



## ● Deterministic where an LLM isn't needed

Booking state and stage transitions are computed in Python.

Conversational turns bypass reasoning and task execution entirely, saving time and cost.

E.g.: greetings; recalling information mentioned earlier in the chat.

## ● Parallelization to reduce latency

Parallelization was used throughout the agent to decrease response time.

E.g.: Translation runs concurrently with intent classification; external location lookup runs concurrently with plan generation.

## ● Guardrails across the whole path

Constraints for security, factual correctness and safe medical-domain behavior sit at every stage, not only in the final prompt.

E.g.: View only Database access for the Query Agent, ID checking and entity validation to ensure entities used in the answer are genuinely in the DB

## Evaluation Design

Once the initial chat version was created a competitive benchmark was designed and run against Gemini and ChatGPT

### 25 scenarios · 8 categories

General consultation, hospital search, pharmacy, emergency, location awareness, products & treatments, booking, robustness

### 4 languages

DB supported languages were tested: Korean, English, Japanese and Chinese. Tests included multi-turn scenarios that must hold context across turns

### Binary rubric

3–5 pass/fail criteria per scenario covering task completion, factual accuracy and safety — no partial credit

### Named verification source

Each criterion checked against a stated source: hospital sites, HIRA, the Seoul pharmacists' 24-hour list, drug-interaction references

### From findings to a schedule

- Failures classified into 9 structural errors plus 1 UI issue
- Each fix mapped to the specific rubric criteria it unblocks
- Expected gain computed per fix, then sequenced by effort
- Each item given an expected completion date
- Same 25 scenarios re-run after the fixes, so the gain is measured rather than assumed

## 07

# Troubleshooting 1 — Location Recognition

## Problem

Landmarks, neighbourhood names and subway stations were not recognised as geographic anchors. A request for dermatology clinics near Gangnam Station searched the whole of Gangnam-gu; a request for 24-hour pharmacies in Myeongdong returned nothing at all.

## Root cause

The TripDoc DB only contained city and district names. Location entities were resolved purely by text-matching against place names stored in the database. Any name absent from that table silently degraded to the enclosing district, or to an empty result.

## Fix

Preloaded a known city/district index into memory as a first-pass gate, then resolved unmatched names to GPS coordinates through an external Google Places API, and searched by proximity from those coordinates. Resolved coordinates are stored in the DB so the same name is cached for future searches. The lookup runs in parallel with the planning and reasoning step.

## Result

Location-aware queries resolve to the intended point rather than the district. Also, the lookup (~250–400 ms) overlaps the reasoning call (~500–800 ms) so the external round-trip is absorbed inside existing latency instead of added on top.

## Troubleshooting 2 — False Negatives in DB Retrieval

### Problem

The assistant reported "not found" for data that existed. A search for an ophthalmologist returned nothing; a request for skincare packages suited to women in their 30s failed despite matching products being in the catalogue.

### Fix

Gave the planner an explicit schema-aware mapping step: user intent is translated into field values the database actually holds before any query is issued. Separately, replaced the bare "not found" response with a generated fallback that offers alternatives and a next step.

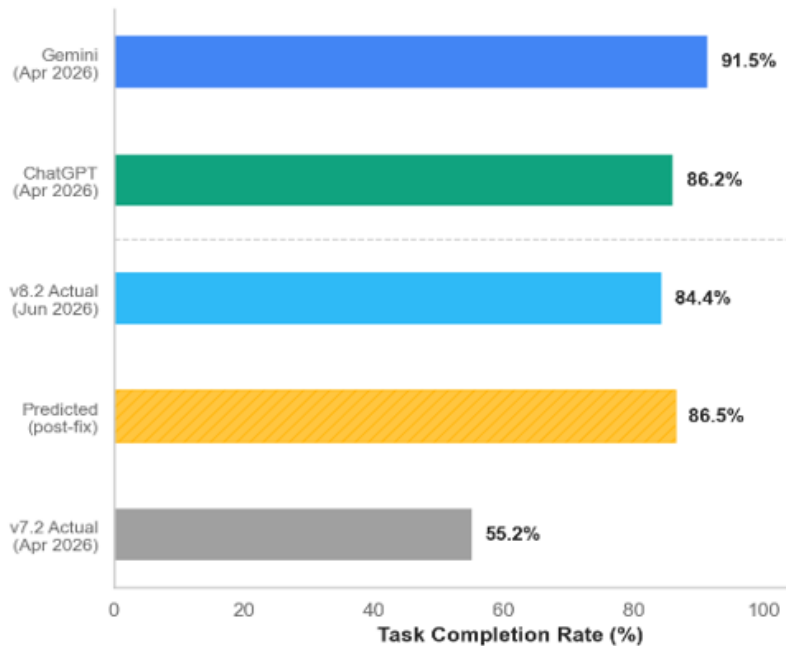
### Root cause

Two distinct faults. The planner passed the user's own wording through as a search term, matching the job title "ophthalmologist" against a specialty field holding "ophthalmology". It framed requests as attributes the schema does not store, such as searching for MRI equipment when only department fields exist.

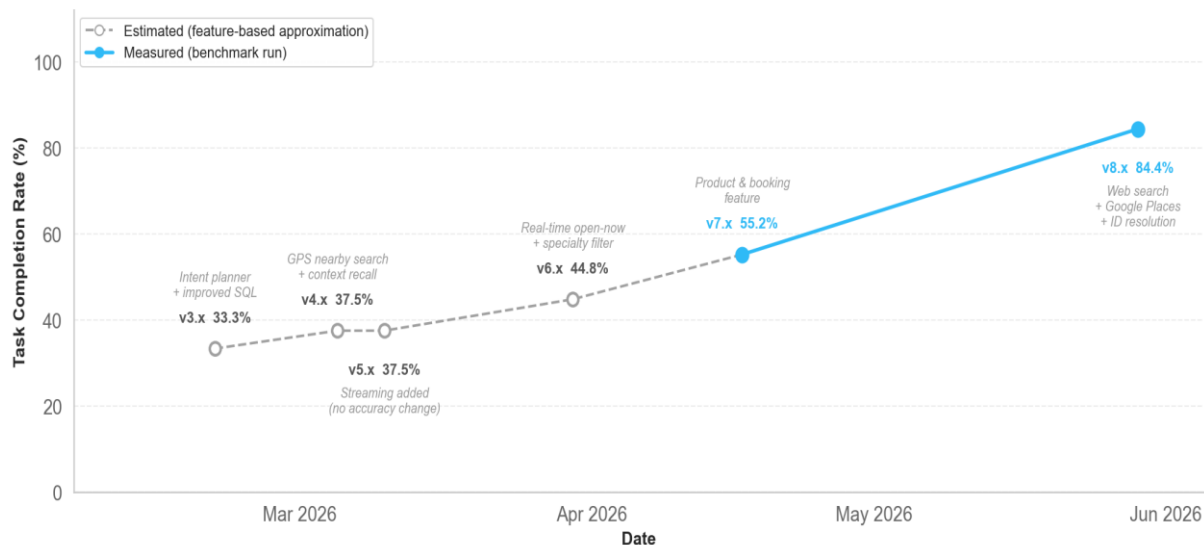
### Result

Hospital search and product/treatment categories both improved. Just as importantly, a miss now ends in a usable answer that the user can follow up on instead of a dead end.

AI Chat Core Feature Performance  
(Tripdoc v7.2 → v8.2 vs Competitors)



AI Chat Core Feature Performance — Version History



## Core Results

Task completion rose from 55.2% to 84.4% across the full rubric, near on par with ChatGPT and within 8 points of Gemini.

The post-fix forecast from the fix plan was 86.5%; measured 84.4% — within 2 points, supporting the failure diagnosis.

## Reading the charts

**Predicted (post-fix)** — a forecast from the fix plan, calculated before re-testing began; not a scored benchmark run.

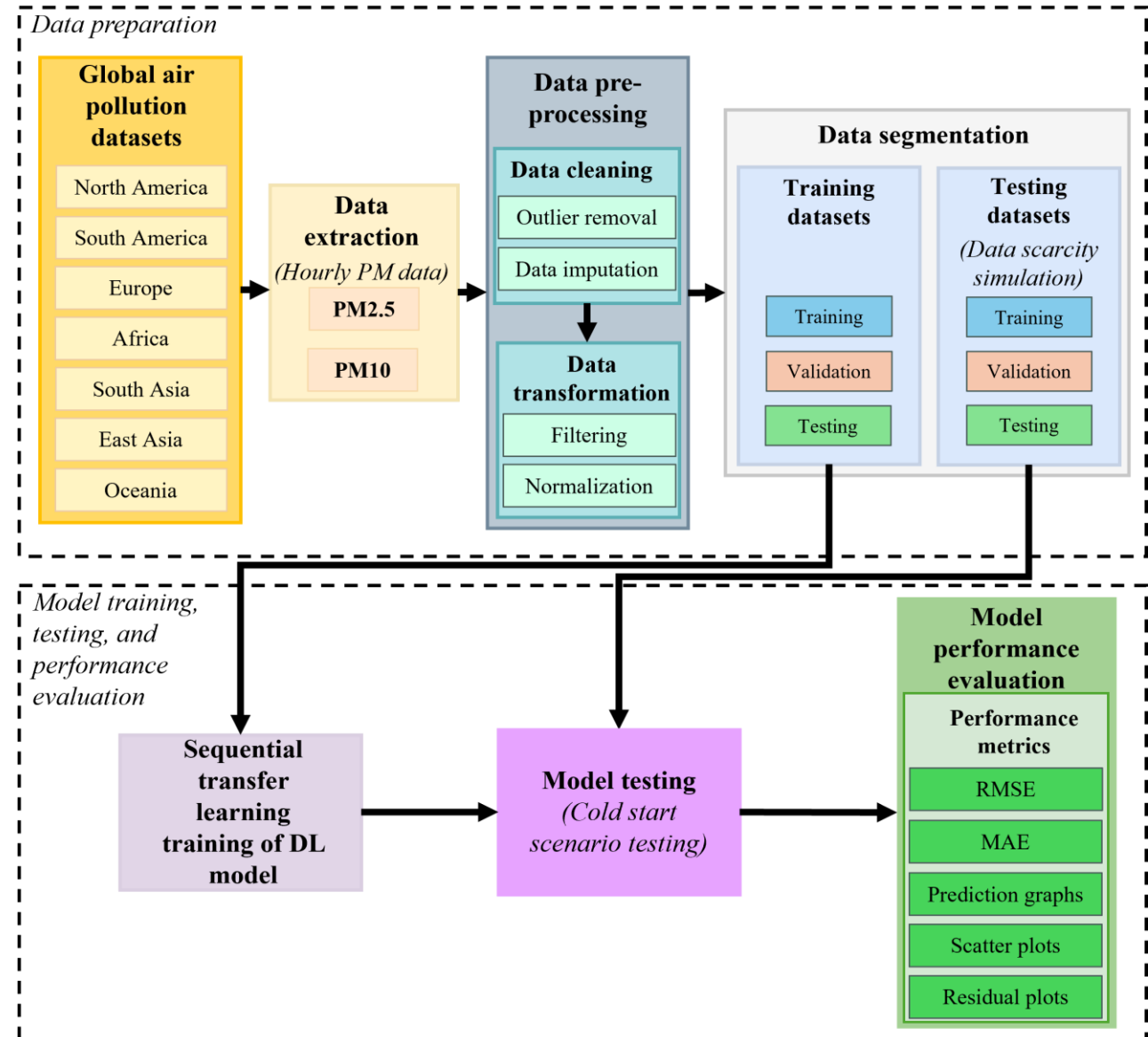
**Estimated (v3–v6)** — full 25-scenario testing started once the system reached feature completeness (v7.2); earlier versions were only estimated due to their limited features

## Problem

Air-quality models are trained per region and usually generalize to other regions poorly. Regions without years of monitoring history are unable to build accurate prediction models. This can be described as a cold-start problem with a public-health cost.

## Method

- Sequential transfer learning: trained on one region at a time, carrying weights forward, across seven world regions (US → Colombia → UK → Uganda → India → China → Australia)
- Preprocessing: STL decomposition for outlier removal, KNN imputation, Butterworth low-pass filtering, min-max normalization
- Cold start simulated by restricting target-region data from 0.5 weeks up to 3 months. Three unseen datasets were used (Korea, Mexico & Spain)
- Three architectures compared: LSTM, 1DCNN, GRU — each against its own non transfer learning trained baseline (trained on target-region data only) with RMSE and MAE used as performance metrics



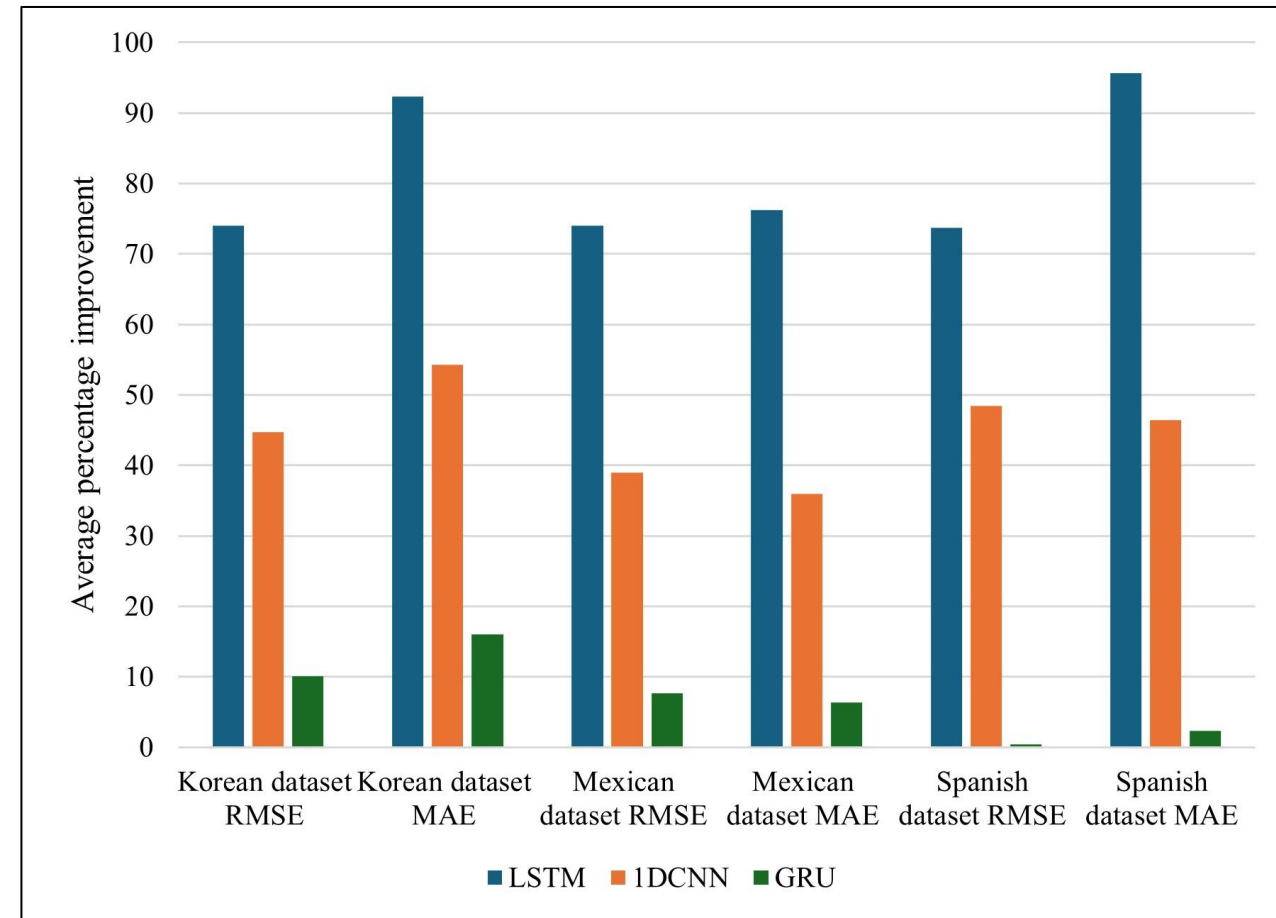
# 11 MSc Research — Results

## Results

- LSTM outperformed its baseline in every data-scarcity scenario tested, reducing forecast error by 73–95% on average and by at least 25% in every scenario.
- Average gains were smaller for the 1DCNN (35–55%) and the GRU (0.4–16%), suggesting the LSTM model is the best candidate for this method
- Model gains were consistent across all three unseen target regions, suggesting learned pollution patterns can transfer between regions
- Error reductions held even with only 0.5 weeks of target-region data, supporting the approach for cold-start forecasting

## Limitations and Future Work

- Hyperparameters tuned by random search; Bayesian optimisation (e.g Optuna) could yield improved performance
- Fixed 50-epoch training instead of early stopping
- Negative transfer appeared in some orderings, incorporating domain selection methods could help reduce this



Average Performance improvement across all tested models in target regions

# 12 Personal Project — Multilingual RAG FAQ Assistant

Retrieval-augmented chatbot for Trinidad & Tobago eVisa FAQs · [github.com/ChrisDanAsh/RAG-LLM-Chatbot](https://github.com/ChrisDanAsh/RAG-LLM-Chatbot) · 2025

## Purpose

The official eVisa FAQ exists only in English, which can be a hinderance to non-English speakers. The project answers visa questions in the user's own language from the official document, so answers are based on the source rather than on model memory.

## Method

- Source PDF chunked at 500 and 1500 characters with 50-character overlap
- Chunks embedded with a HuggingFace sentence-transformer; retrieval by semantic similarity
- Gemini generates the answer from the retrieved chunks only
- Non-English queries are detected and translated, then routed through the same pipeline since the corpus and embedding model are English-only.

## Future Work

Vector store is rebuilt in memory at every start, and the interface is a CLI. Next: persist the index and serve it behind FastAPI in a container.

```
(.venv) @ChrisDanAsh →/workspaces/LLM-Chatbot/llm-chatbot (v2/FAQ-chatbot) $ python -m agent

Initialization complete. You can start chatting now.

Type 'exit' or 'quit' to stop.

You: Who can apply for a Visa?
**Who can apply for a Visa online?**

To determine your eligibility for an online visa application, you will be asked a series of questions during the application process. If you meet the criteria, you can proceed to complete the online form.

You: 비자를 신청하려면 어떻게 해야 하나요?
온라인으로 비자를 신청하려면 먼저 "사진 확인" 도구를 사용하여 사진이 적합한지 확인해야 합니다. 그런 다음 "비자 신청"을 선택하고 디지털 사진 및 지원 서류를 제출하여 온라인 신청서를 작성합니다. 처리 수수료를 지불하기 위한 신용카드도 필요합니다.

You: ¿Qué tipo de foto necesito para la Visa?
Tu foto debe estar en formato JPEG, PNG o BMP. Debe tener un tamaño entre 15 KB y 7584 KB, y sus dimensiones deben oscilar entre 480 píxeles de ancho por 640 píxeles de alto y 4800 píxeles de ancho por 6400 píxeles de alto. Puedes usar la herramienta "Verifica tu foto" en la página de inicio de eVisa para asegurarte de que cumple con los requisitos.

You: Comment puis-je payer ma carte Visa?
Vous pouvez payer votre demande de visa électronique TTO par carte de crédit sur un site web sécurisé tiers pendant le processus de candidature. Une fois le paiement effectué, vous recevrez une confirmation.

You: exit
Goodbye!
```

## Results

20 FAQ based questions (17 in scope, 3 out of scope) used to test the performance of the retrieval pipeline

| Metric   | 500 chunk size | 1500 chunk size |
|----------|----------------|-----------------|
| recall@1 | 0.76           | 0.76            |
| recall@3 | 0.94           | <b>0.88</b>     |
| recall@4 | 0.94           | <b>0.94</b>     |

- Based on the results 1500 chunk size at retrieval\_k = 4 was selected since it was above the size of the longest page (1013 chars), keeping matched data intact
- The benefit of this choice is not measured by recall@k but should improve the accuracy of the answers at the cost of slightly higher latency and prompt length

# 13 R&D Engineer Project — Autonomous Disinfection Robot

Government funded Disinfection Robot project for Trinidad & Tobago · University of the West Indies · 2020 - 2021

## Purpose

During the COVID-19 pandemic, the government of Trinidad and Tobago sought a solution for cleaning hospital rooms.

## Roles

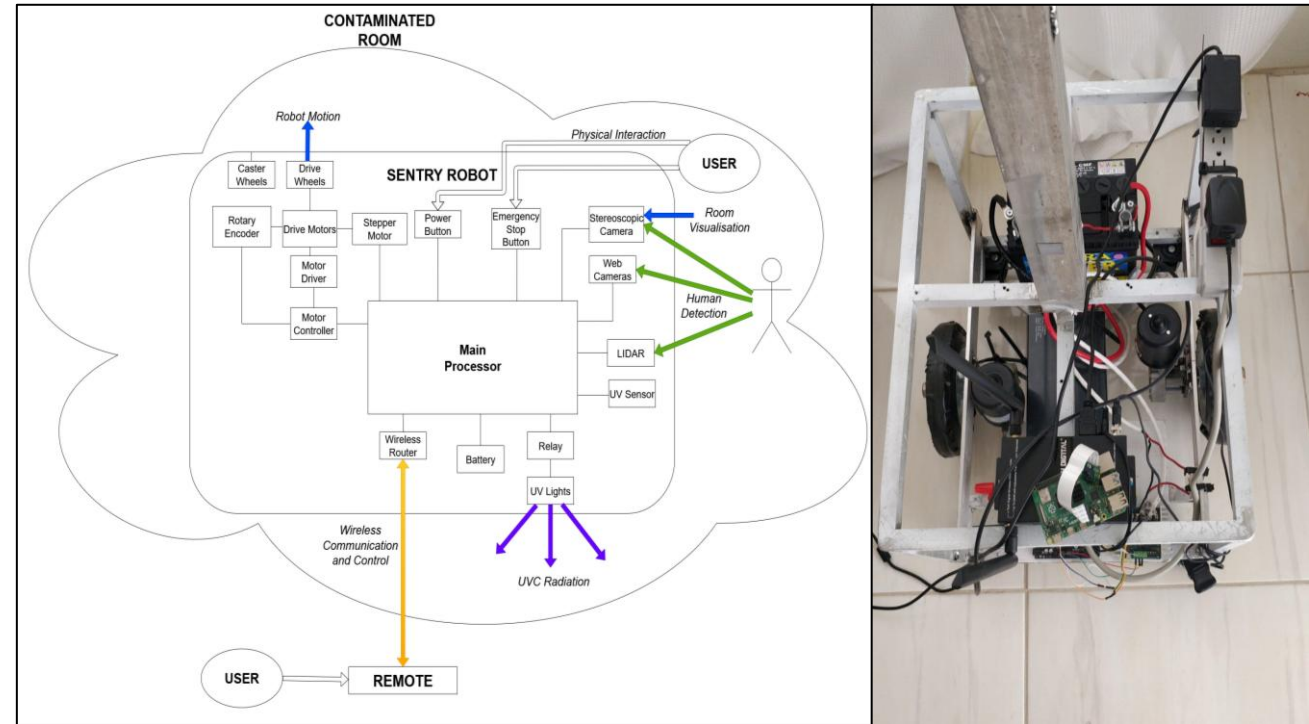
- Project Manager Assistant: Robot system architecture design
- Embedded Systems Engineer: Motor controller implementation

## Method

- Arduino was used for the motor controller to simultaneously control each motor.
- PID control was compared against PD control with randomly searched gains; PD produced better results and was kept
- Navigation commands were transmitted from the main system controller via UART

## Problem

- Motor encoder ticks were being missed, so the control loop was acting on incorrect velocity readings
- The Arduino Mega's sampling rate was too slow relative to the encoder tick frequency



(1) Disinfection Robot System High Level Architecture Diagram (2) Robot Prototype

## Fix + Results

- Migrated to an Arduino Due (84 MHz vs the Mega's 16 MHz), which captured ticks reliably
- The Due runs 3.3 V logic where the Mega ran 5 V, so encoder and communication lines had to be level-shifted.
- This allowed for stable simultaneous control of both drive motors

## 14 Other Projects

### ● Construction Site Hole Detection

Python · PyTorch · YOLOv5

- Prototyped hazard detection system for construction sites
- Applied customized hole detection YOLOv5 model to construction site images

### ● Air Disinfection Unit

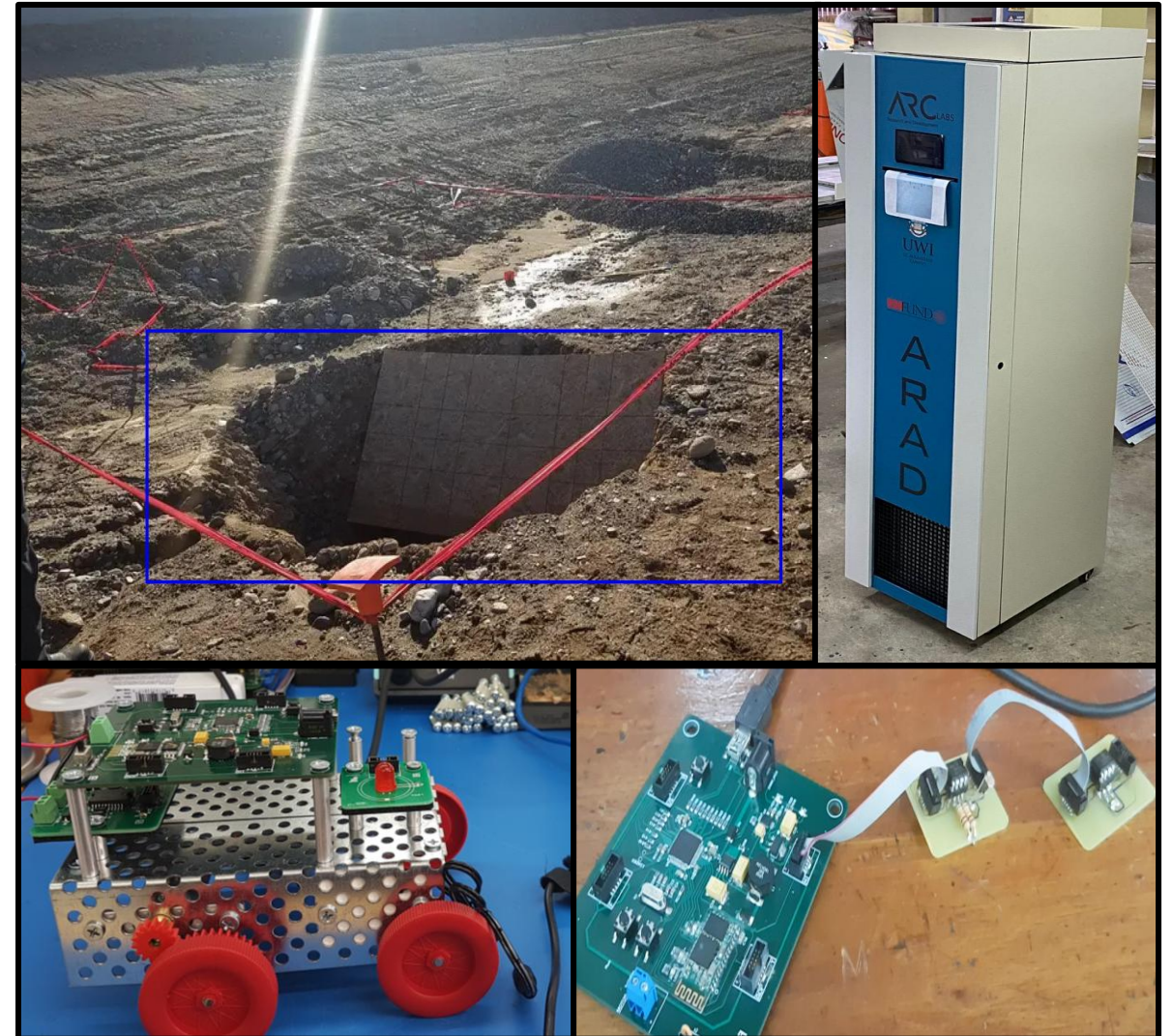
Python · Raspberry Pi

Designed and implemented air pollution sensor and actuator control system

### ● Educational Robotics Kit

C# · C++ · C · PIC · PCB design

- Firmware and programming environment for a teaching kit
- Created custom Daisy Chainable PCB peripherals using I2C communication protocol



(1) Construction site hole detection (2) Air Disinfection Unit (3) Educational Robotics Kit (4) Custom PCB peripherals

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# How I Work

- **In-depth Design before Build**

Architecture and component design are thoroughly thought out and documented before implementation, then used as the reference the build is checked against

- **Diagnose Root Causes, not Symptoms**

Symptoms are traced to the layer that produced them; schema, hardware, or metric, rather than patched where they appear

- **From Findings to a Scheduled Plan**

Test results converted into structured findings, then a prioritized, dated plan; what to fix, in what order, and why

# Thank You

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